

In-Class Quiz: May 7

Points: 10/10

✓ **Correct** 1/1 Points

1. What is the primary advantage of pipelining in a processor? 

- ☐ It reduces the execution time of a single isolated instruction.
- ☒ It keeps all parts of the data path busy all the time.
- ☐ It eliminates the need for an Execute (EX) unit.
- ☐ It allows multiple instructions to write to the same register simultaneously.

✓ **Correct** 1/1 Points

2. In a perfectly balanced 5-stage pipeline, if an instruction takes T seconds to complete without pipelining, how often does a new instruction complete its execution with pipelining?



- ☐ Every T seconds
- ☐ Every 5T seconds
- ☒ Every T/5 seconds
- ☐ Every T^5 seconds

✓ **Correct** 1/1 Points

3. In the 5-stage pipeline design, what component is inserted between consecutive stages and is triggered by a negative clock edge? 

- ☐ Multiplexers
- ☒ Latches (Pipeline Registers)
- ☐ Forwarding Units
- ☐ Hardware Interlocks

✓ **Correct** 1/1 Points

4. Which of the following data hazards is the ONLY one possible in a strictly in-order pipeline?



- ☐ Write After Write (WAW)
- ☐ Write After Read (WAR)
- ☐ Read After Read (RAR)
- ☒ Read After Write (RAW)

✓ **Correct** 1/1 Points

5. What specifically causes a Structural Hazard in a pipelined processor? 

- ☒ Two instructions have a conflict on the same set of resources in a given cycle.
- ☐ Instructions from the wrong path of a branch are fetched.
- ☐ An instruction requires data that has not yet been computed.
- ☐ A hardware interrupt forces the pipeline to flush.

✓ **Correct** 1/1 Points

6. To solve Control Hazards in software, compilers can utilize a technique involving "delay slots." What does this entail? 

- ☐ Disabling the clock for a few cycles until the branch resolves.

- ☒ Moving valid instructions that must execute anyway into the spaces immediately following a branch instruction.
- ☐ Flushing the entire register file when a branch is detected.
- ☐ Inserting a "Data-Lock" command into the code stream.

✓ **Correct** 1/1 Points

7. When implementing a hardware "Data-Lock" interlock to resolve a RAW hazard, which specific pipeline stages must be stalled? 

- ☐ EX and MA
- ☐ MA and RW
- ☒ IF and OF
- ☐ OF and EX

✓ **Correct** 1/1 Points

8. What exactly constitutes a "pipeline bubble" inserted by hardware to handle stalls? 

- ☐ A temporary hardware buffer.
- ☒ A nop (no-operation) instruction packet.
- ☐ A dedicated signal that speeds up the system clock.
- ☐ A mechanism to clear data memory.

✓ **Correct** 1/1 Points

9. What is the primary benefit of adding data "forwarding" paths to the pipeline? 

- ☐ It completely eliminates the need for an Instruction Fetch unit.

- ☒ It passes a needed value from a later stage directly to an earlier stage, reducing or eliminating stalls.
- ☐ It ensures that instructions are always executed out-of-order.
- ☐ It acts as an interlock that stops the clock entirely.

✓ **Correct** 1/1 Points

10. According to the rules for creating forwarding paths, which of the following pipeline stages is NEVER a part of any forwarding path? 

- ☐ Operand Fetch (OF)
- ☐ Memory Access (MA)
- ☐ Execute (EX)
- ☒ Instruction Fetch (IF)



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